Rebecca A. Deek

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Education

2018-2023 PhD in Biostatistics — Philadelphia, PA
University of Pennsylvania, Perelman School of Medicine
Advisor: Hongzhe Li
Committee: Mingyao Li (Chair), Jing Huang, and Ronald G. Collman

2016-2018 MS in Biostatistics — New York, NY
Columbia University, Mailman School of Public Health

2013-2016 BS in Biological Sciences — Newark, NJ
New Jersey Institute of Technology, Albert Dorman Honors College
Rutgers University

Professional appointments

2023-Present Assistant Professor — Pittsburgh, PA
Department of Biostatistics and Health Data Science
School of Public Health
University of Pittsburgh

2018-2023 Graduate Research Assistant — Philadelphia, PA
Department of Biostatistics, Epidemiology, & Informatics
Perelman School of Medicine
University of Pennsylvania

Honors & awards

2026 Nominated for the Dr. James L. Craig Endowed Excellence in Graduate Education Award
University of Pittsburgh School of Public Health

2026 Early Career Paper Award
American Statistical Association – Biometrics Section, JSM 2026

2022 Best Poster Award
Penn Conference on Big Data in Biomedical and Population Health Sciences

2016-2018 Hu Family Scholarship
Columbia University

2016 President's Medal for Academic Excellence
New Jersey Institute of Technology

2016 Outstanding Honors Research Award
NJIT Albert Dorman Honors College

2015 Leadership Service Award
NJIT Albert Dorman Honors College

2013-2016 Full Merit Scholarship
NJIT Albert Dorman Honors College

Funding

Active

2026-2028 UPMC Competitive Medical Research Fund: "Novel methods for dimension reduction in microbiome and multiomics studies"
Role: Principal Investigator (PI: Deek RA)

2026-2030 NIH UC2AR086243: "Whole person experience of pain: Novel integration with low back pain phenotypes"
Role: Co-investigator, 15% FTE. (PI: Sowa G and Vo N)

2026-2030 NIH R01AG092661: "Multiscale Models of Age-Specific Neurometabolic Coupling"
Role: Co-investigator, 10% FTE. (PI: Iordanova B and Zhan L)

2025-2026 University of Pittsburgh Trans-disciplinary Collaboration Pilot Award: "Identifying inflammatory biomarkers of dementia via multimodal integration of neuroimaging & omics data"
Role: Principal Investigator (MPI: Deek RA and Iordanova B)

2025-2030 NIH P30AG066468: "Alzheimer's Disease Research Center"
Role: Co-investigator, 5% FTE. (PI: Lopez O)

2024-2027 MWRI Magee Prize: "EGFL6 - A link between obesity, high-risk endometrial cancer, and racial disparities"
Role: Co-investigator, 8% FTE. (PI: Buckanovich R)

2023-2028 NIH R01AG083874: "Plasma brain-derived tau: A novel Alzheimer's disease-type neurodegeneration biomarker with potential to complete the AT(N) scheme in blood"
Role: Co-investigator, 20% FTE. (PI: Karikari T)

2023-2028 NIH P50CA272218: "HCC Ovarian Cancer SPORE"
Role: Co-investigator, 12% FTE. (MPI: Buckanovich R and Edwards R)

Publications

∗ Advisee/mentee
† Co-first author
§ Co-senior author

Statistical methodology

1. **Zang C**∗, **Deek RA**§, and Tang L§. Supervised matrix factorization for estimating individualized treatment rule: Incorporating medical notes in decision making. *Statistical Analysis and Data Mining (In press)*.[1]
2. **Zang C**∗, Tang L, and **Deek RA**. Supervised and covariate-adjusted dimension reduction for prediction and visualization using microbiome data. *Briefings in Bioinformatics* 27, no. 5 (2026): bbag590.[2]

[1] A prior version of this paper won the 2024 Miheala Serban Award for Best Poster Presentation from the ASA's Pittsburgh Chapter and JSM 2025 Best Poster Award from the ASA Section on Text Analysis.

[2] A prior version of this paper received a JSM 2026 Student Paper Honorable Mention from the ASA Mental Health Statistics Section.

3. Zhang VZ[†], Ferreira PCL[†], Dong Y, Minhas D, Povala G, Bellaver B, Alzheimer's Disease Neuroimaging Initiative, Pascoal TA, Zeng X, Karikari TK, Cohen AD, **Deek RA**, Wu Q[§], and Tudorascu DL[§]. Evaluation of harmonization methods to mitigate assay and cohort effects in plasma p-tau217. *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring* 18, no. 4 (2026): e70518.
4. Yan B, Nam Y, Li L, **Deek RA**, Li H, and Ma S. Recent advances in deep learning and language models for studying the microbiome. *Frontiers in Genetics* 15 (2025): 1494474.
5. Little A, **Deek RA**, Zhang A, Zhao N, Ling W, and Wu MC. Enhanced visualization of microbiome data in repeated measures designs. *Frontiers in Genetics* 15 (2024): 1480972.
6. **Deek RA**, Ma S, Lewis J, and Li H. Statistical and computational methods for integrating microbiome, host genomics, and metabolomics data. *eLife* 13 (2024): e88956.
7. **Deek RA** and Li H. Inference of microbial covariation networks using copula models with mixture margins. *Bioinformatics* 39, no. 7 (2023): btad413.
8. **Deek RA** and Li H. A zero-inflated latent dirichlet allocation model for microbiome studies. *Frontiers in Genetics* 11 (2021): 602594.
9. Chen J[†], King E[†], **Deek R**[†], Wei Z, Yu Y, Grill D, and Ballman K. An omnibus test for differential distribution analysis of microbiome sequencing data. *Bioinformatics* 34, no. 4 (2018): 643–651.

Collaborative

10. Schweitzer N, Shen Y, Zhao Y, Cover CG, Alam MS, **Deek RA**, Aizenstein HJ, Wu M, Koldamova R, Vazquez AL, Fitz NF, and Iordanova B. Small vessel tortuosity is an early, cross-species biomarker of dysfunctional mural cell contractility and aberrant angiogenesis in Alzheimer's disease. *Alzheimer's & Dementia (In press)*.
11. Carlson KM, Owoc MS, Nilsen ML, Mowery YM, **Deek RA**, Contrera KJ, Hetrick J, Osazuwa-Peters N, Zandberg DP, Stinnett S, Johnson JT, and Mazul AL. Clinical and social determinants of sustained head and neck cancer survivorship clinic attendance. *Journal of Cancer Survivorship* (2026).
12. **Deek RA**[†], Balogun WG[†], Zeng X, Triana-Baltzer G, Pascoal TA, Kolb HC, Snitz B, Cohen AD, and Karikari TK. Head-to-head comparison of plasma p-tau217 immunoassays for incipient Alzheimer's disease in community cohorts. *Alzheimer's Research & Therapy* (2026).
13. Brodman ST, **Heaton N**, Triana-Baltzer G, Zeng X, Gogola A, Kamboh MI, Villemagne VL, Lopez OL, Kolb H, **Deek RA**, Cohen AD[§], and Karikari TK[§]. Plasma biomarkers, brain amyloid-beta pathology, and cortical thickness in a non-Hispanic White and Black/African American middle-aged community cohort: The HCP-CoBRA study. *Alzheimer's & Dementia* 21, no. 12 (2025): e70985.
14. Whiteside SA, McGinniss JE, **Deek RA**, Merenstein C, Britton N, Simon-Soro A, Oyster M, Kalman L, Brown MC, Graham-Wooten J, McDyer JF, Shah P, D'Alessio F, Cantu E, Clausen ES, Li H, Diamond JM, Bushman FD, Christie JD, and Collman RG. Lung transplant for CF: Low lung bacterial burden and immune mediators in year one associate with clad development. *Journal of Cystic Fibrosis* (2025): S1569199325024865.
15. **Krsmanovic G***, **Deek RA**[§], and Deek FP[§]. The Development and Maintenance of Thinking Processes and Learning Abilities in the Era of Artificial Intelligence. *Computer* 58, no. 9 (2025): 80–90.
16. **Deek RA**, Kim J, Kumar R, Medina B, Langenfeld J, Kontopidis I, Lazzaro R, Nie K, Hathout L, Guler O, Onal C, Patel E, Haigentz M, Jabbour SK, and Deek MP. Clinical and Genomic Features and Prognostic Biomarkers of Oligometastatic Nonsmall Cell Lung Cancer. *Clinical Lung Cancer* (2025): S1525730425001536.

17. Carlson KM, Abdul-Rahman N.-H, **Deek RA**, Cenname A, Cai W, Demanelis K, Zandberg DP, Contrera KJ, Zevallos JP, Martinson J, and Mazul AL. Trends in Human Papillomavirus Testing Among Patients With Oropharyngeal Cancer. *JAMA Network Open* 8, no. 7 (2025): e2523917.
18. Sehrawat A, Zeng X, Abrahamson EE, **Deek RA**, Gogola A, Kamboh MI, Pascoal TA, Villemagne VL, Lopez OL, Ikonomovic MD, Snitz BE, Cohen AD, and Karikari TK. Pittsburgh plasma p-tau217: Classification accuracies for autosomal dominant and sporadic Alzheimer's disease in the community. *Alzheimer's & Dementia* 21, no. 7 (2025): e70409.
19. Duan H, Ren J, Wei S, Yang Z, Li C, Wang Z, Li M, Wei Z, Liu Y, Wang X, Lan H, Zeng Z, Xie M, Xie Y, Wu S, Hu W, Guo C, Zhang X, Liang L, Yu C, Mou Y, Jiang Y, Li H, Sugarman E, **Deek RA**, Chen Z, Li T, Chen Y, Yao M, Chen L, Liu L, Zhang G, and Mou Y. Integrated analyses of multi-omic data derived from paired primary lung cancer and brain metastasis reveal the metabolic vulnerability as a novel therapeutic target. *Genome Medicine* 16, no. 1 (2024): 138.
20. Wang T, Kim J, Kumar R, **Deek RA**, Stephenson R, Mayer T, Saraiya B, Ghodoussipour S, Jang T, Golombos D, Packiam V, Ennis R, Hathout L, Jabbour SK, Guler O, Onal C, and Deek MP. Landscape and prognostic significance of oncogene drivers in metastatic castration sensitive prostate cancer. *Translational Cancer Research* 13, no. 11 (2024): 6235–6245.
21. Sutera P, Kim J, Kumar R, **Deek RA**, Stephenson R, Mayer T, Saraiya B, Ghodoussipour S, Jang T, Golombos D, Packiam V, Ennis R, Hathout L, Jabbour SK, Guler O, Onal C, Tran PT, and Deek MP. *PIK3/Akt/mTOR* pathway alterations in metastatic castration-sensitive prostate cancer. *The Prostate* 84, no. 14 (2024): 1301–1308.
22. Sutera P†, Deek MP†, **Deek RA**, Guler OC, Hurmuz P, Reyhan M, Rowe S, Radwan N, Dipasquale S, Hrinivich WT, Lowe K, Ren L, Saraiya B, Ennis R, Hathout L, Mayer T, Deweese TL, Song DY, Kiess A, Oymak E, Pienta K, Feng F, Pomper M, Ozyigit G, Tran PT, Onal C, and Phillips RM. Prostate-Specific Membrane Antigen PET Response Associates with Metastasis-Free Survival After Stereotactic Ablative Radiation in Oligometastatic Prostate Cancer. *Advances in Radiation Oncology* 9, no. 7 (2024): 101507.
23. Deek AJ, **Deek RA**, Clegg DJ, Blackburn C, Holley T, and Gotcher JE. To what extent do oral and maxillofacial surgery residents suffer from imposter syndrome? *Oral Surgery, Oral Medicine, Oral Pathology and Oral Radiology* 138, no. 1 (2024): 21–27.
24. Deek AJ, Clegg DJ, **Deek RA**, Blackburn C, Scott CA, Holley T, and Gotcher JE. Risk Factors for Pediatric Facial Trauma in an Appalachian Region: An Epidemiological Review at a Single Institution. *Journal of Oral and Maxillofacial Surgery* 82, no. 3 (2024): 306–313.
25. Khullar K, **Deek RA**, Nelson B, Gaines D, Corrigan KL, LeCompte MC, Deville C, Deek MP, and Jabbour SK. Gender and the Receipt of the Association of Residents in Radiation Oncology Educator of the Year Award. *Advances in Radiation Oncology* 9, no. 1 (2024): 101305.
26. McGinniss JE, Whiteside SA, **Deek RA**, Simon-Soro A, Graham-Wooten J, Oyster M, Brown MD, Cantu E, Diamond JM, Li H, Christie JD, Bushman FD, and Collman RG. The Lung Allograft Microbiome Associates with Pepsin, Inflammation, and Primary Graft Dysfunction. *American Journal of Respiratory and Critical Care Medicine* 206, no. 12 (2022): 1508–1521.
27. Deek MP†, Van Der Eecken K†, Sutera P, **Deek RA**, Fonteyne V, Mendes AA, Decaestecker K, Kiess AP, Lumen N, Phillips R, De Bruycker A, Mishra M, Rana Z, Molitoris J, Lambert B, Delrue L, Wang H, Lowe K, Verbeke S, Van Dorpe J, Bultijnck R, Villeirs G, De Man K, Ameye F, Song DY, DeWeese T, Paller CJ, Feng FY, Wyatt A, Pienta KJ, Diehn M, Bentzen SM, Joniau S, Vanhaverbeke F, De Meerleer G, Antonarakis ES, Lotan TL, Berlin A, Siva S, Ost P, and Tran PT. Long-Term Outcomes and Genetic Predictors of Response to Metastasis-Directed Therapy Versus Observation in Oligometastatic Prostate Cancer: Analysis of STOMP and ORIOLE Trials. *Journal of Clinical Oncology* 40, no. 29 (2022): 3377–3382.

28. Wei S, Yin D, Yu S, Lin X, Savani MR, Du K, Ku Y, Wu D, Li S, Liu H, Tian M, Chen Y, Bowie M, Hariharan S, Waitkus M, Keir ST, Sugarman ET, **Deek RA**, Labrie M, Khasraw M, Lu Y, Mills GB, Herlyn M, Wu K, Liu L, Wei Z, Flaherty KT, Abdullah K, Zhang G, and Ashley DM. Antitumor Activity of a Mitochondrial-Targeted HSP90 Inhibitor in Gliomas. *Clinical Cancer Research* 28, no. 10 (2022): 2180–2195.
29. **Deek R**†, Nasser J†, Ghanem A†, Mardelli M†, Khazen G, Salloum AK, Abchee A, Ghassibe-Sabbagh M, and Zalloua P. Genome-wide association analysis of HDL-C in a Lebanese cohort. *PLOS ONE* 14, no. 6 (2019): e0218443.

Published proceedings and columns

30. **Tommarello ND*** and **Deek RA**. (2024). The Convergence of the Internet of Things and Artificial Intelligence in Medicine: Assessing the Benefits, Challenges, and Risks. *Computer*, 57(2), 95-99.
31. **Deek R**. (2015). Lessons From Nature: Defensive Designs for the Built Environment. *In 5th International Conference on Building Resilience*.

Under review or revision and preprints

Statistical Methodology

32. **Liu T***, Zeng X, Snitz BE, Karikari TK, **Deek RA**. The dangers of data double dipping in assessing the classification accuracies of blood biomarkers in Alzheimer's disease and related disorder research. *Invited revision*.
33. **Deek RA**. Distance-based mutual information test of multivariate association in microbiome multiomics studies. *Invited revision*.[3]
34. **Lin W***,†, **Zhao K***,†, Demirci FY, Sweet RA, Wang J§, **Deek RA**§. Inferring cell-type-specific networks from single-cell epigenomic data. *Invited revision*.
35. **Deek RA**†, Xue F†, Lewis JD, Li H. Estimation and false discovery control of multivariate causal effects in omics studies. *Invited revision*.
36. **Heaton N***, Zeng X, Snitz BE, Karikari TK, **Deek RA**. Comparative evaluation of machine learning approaches for blood biomarker prediction of amyloid-beta pathology, tau tangles and neurodegeneration in the community. *Under review*.
37. Sun J, Morrison R, Zhenjiang F, Sriram A, Zarour H, Rodrigues R, Trinchieri G, **Deek RA**, Park HJ. DeepMicroNET: Disentangling Trait Specific Biology From Overlapping Signals in Human Biomedical Data. *Under review*.
38. Sun J, Fan Z, Sriram A, **Deek RA**, Kim S, Park HJ. Unveiling the gut microbiome network fingerprint: A deep learning approach to predict immunotherapy response in melanoma patients. *In revision*.
39. **Deek RA** and Li H. Mixture margin random-effects copula models for inferring temporally conserved microbial covariation networks from longitudinal data. *In revision*.

Collaborative

40. Grizzanti J, Zhang Y, **Deek RA**, Lafferty TK, Jacobsen E, Bellaver B, Ferreira PCL, Zeng X, Chang CH, Pascoal TA, Snitz BE, Villemagne VL, Ganguli M, Karikari TK. Blood markers of neurodegeneration: Sex differences and associations with glycemic control and kidney function in a population-based cohort. *Invited revision*.

[3] This paper won a JSM 2026 Early Career Paper Award from the ASA Biometrics Section.

41. Yang TJ, Guler OC, **Deek RA**, Shah S, Narra LR, Sutera P, Saraiya B, Mayer T, Stephenson R, Hathout L, Ennis R, Parikh R, Mattes M, Torun N, Elmali A, Demirhan B, Elsamra S, Packiam VT, Ghodoussipour S, Golombos DM, Jang T, Jabbour SK, Tran PT, Nie K, Onal C, Deek MP, Matched Pre- and Post-Treatment PSMA PET Imaging Biomarkers of Response in Intermediate and High Risk Prostate Cancer Treated with Definitive Radiation and Androgen Deprivation Therapy. *Invited revision*.
42. Zeng X, **Heaton N***, **Deek RA**, Nafash MN, Farinas MF, Gu J, Choity LT, Balogun WG, Shi R, Bedison MA, Lopresti BJ, Matan C, Nadkarni NK, Gogola A, Shaaban CE, Aizenstein HJ, Tudorascu DL, Pascoalı TA, Villemagne VL, Berman SB, Ikonomovic MD, Sweet RA, Kofler JK, Snitz BE, Kamboh MI, Cohen AD, Lopez OL, Karikari TK. Plasma biomarkers and incident clinical stage progression in the specialized memory clinic: a longitudinal study. *Under review*.
43. Narra LR, Kumar R, Zhang C, Sutera PA, **Deek RA**, Saraiya B, Mayer T, Stephenson R, Hathout L, Ennis R, Packiam V, Ghodoussipour S, Golombos DM, Jang T, Jabbour S, Parikh R, Tran PT, Nie K, Deek MP. Multi-omics Modeling Predicts for Lesion-Level Prostate Specific Membrane Antigen Avidity in Metastatic Castrate Sensitive Prostate Cancer. *Under review*.

Software

1. dMI: An R package for multivariate global association testing in multiomics studies using distance-based mutual information. Creator and core developer.
2. scoNet: An R package modeling zero-one-inflated single-cell epigenomic data for co-methylation network analysis using copula models with mixture margins.
3. scaPCoA: An R package for supervised and covariate adjusted dimension reduction for microbiome data.
4. CoMiCoN: An R package for two-stage inference and network analysis of bivariate mixture margin copula models. Creator and core developer.
5. zinLDA: An R package for MCMC implementation of zero-inflated latent Dirichlet allocation models for microbiome studies. Creator and core developer.

Presentations

Invited presentations

1. *Distance-based mutual information test of multivariate association in microbiome multiomics studies*. JSM 2026. Boston, MA, USA. August 2026.
2. *Testing for multivariate association in microbiome multiomics with distance-based mutual information*. ISI Regional Statistics Conference. Valletta, Malta. June 2026.
3. *Testing for multivariate association in microbiome multiomics with distance-based mutual information*. STATGEN 2026. Atlanta, GA, USA. May 2026.
4. *Supervised dimension reduction for visualization and prediction with microbiome data*. CMStatistics 2025. London, United Kingdom. December 2025.
5. *Large language models for molecular biology*. 16th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB 2025). Philadelphia, PA, USA. October 2025.
6. *Benchmarking and improving statistical methods for microbiome multiomics integration.* BIRS Workshop on Novel Statistical Approaches for Studying Multi-omics Data. Banff, Canada. July 2025.
7. *Supervised dimensionality reduction and visualization of microbiome data.* ENAR 2025. New Orleans, LA, USA. March 2025.

8. *Joint estimation and false discovery control of causal effects in metabolomics randomized trials.* CMStatistics 2024. London, United Kingdom. December 2024.
9. *A copula model framework for estimating microbial covariation networks.* JSM 2024. Portland, OR, USA. August 2024.
10. *Estimating temporally conserved microbial covariation networks from longitudinal data.* STATGEN 2024. Pittsburgh, PA, USA. May 2024.
11. *Statistical and computational methods for integrating microbiome and host omics data.* CMStatistics 2023. Berlin, Germany. December 2023.
12. *Statistical and computational methods for integrating microbiome and host omics data.* Women in Statistics and Data Science 2023. Bellevue, WA, USA. October 2023.
13. *Statistical and computational methods for integrating microbiome and host omics data.* JSM 2023. Toronto, ON, Canada. August 2023.
14. *Microbial covariation discovery using mixture margin copula models.* IMS 2022. London, UK. June 2022.
15. *Methods for inference of microbial interactions using copula models with mixture margins.* JSM 2020. Philadelphia, PA, USA (Virtual due to COVID-19). August 2020.

Invited seminars

1. *Improving visualization and prediction with microbiome data through supervised dimension reduction*. Department of Human Genetics Seminar. University of Pittsburgh. Pittsburgh, PA, USA. December 2025.
2. *Modeling complex dependency and network structures in zero-inflated metagenomic sequencing data.* Institute for Healthcare Delivery Science Seminar. Icahn School of Medicine at Mount Sinai. New York, NY, USA. November 2024.
3. *Mixture margin copula models for inferring covariation networks with applications in metagenomics.* Department of Statistics Seminar. University of Pittsburgh. Pittsburgh, PA, USA. September 2024.
4. *Advances in statistical methods for metagenomic sequencing data analysis.* Basic and Translational Research in Lung Diseases Seminar. Department of Medicine. University of Pittsburgh. Pittsburgh, PA, USA. April 2024.
5. *Advances in statistical methods for metagenomic sequencing data analysis.* Department of Orthopedics Seminar. University of Pittsburgh. Pittsburgh, PA, USA. March 2024.
6. *Mixture margin copula models for inferring covariation networks with applications in microbiome and single cell genomics*. Department of Statistics. University of Georgia. Athens, GA, USA. January 2023.
7. *Mixture margin copula models for inferring covariation networks with applications in microbiome and single cell genomics*. Department of Biostatistics. Rutgers University. New Brunswick, NJ, USA (Virtual). January 2023.
8. *Mixture margin copula models for inferring covariation networks with applications in microbiome and single cell genomics*. Department of Biostatistics. University of Pittsburgh. Pittsburgh, PA, USA. December 2022.
9. *Mixture margin copula models for inferring covariation networks with applications in microbiome and single cell genomics*. Department of Mathematics and Statistics. The College of New Jersey. Ewing, NJ, USA. December 2022.
10. *Inferring microbial cooccurrence from sequencing data.* NIEHS BCBB Seminar. Research Triangle, NC, USA (Virtual). April 2022.

Contributed Presentations

1. *Statistical and computational approaches for estimating microbial covariation networks.* Rust Belt Microbiome 2024. Pittsburgh, PA, USA. November 2024.
2. *Head-to-head comparison of four different plasma p-tau217 assays to detect brain amyloid pathology and cognitive decline in diverse population-based cohorts.* AAIC 2024. Philadelphia, PA, USA. August 2024.
3. *Inference for conserved microbial covariations using mixture margin random-effects copula models.* Penn Conference on Big Data in Biomedical and Population Health Sciences. Philadelphia, PA, USA. September 2022. (Best Poster Award).
4. *Statistical methods for microbiome driven integration of omics data.* ENAR 2023. Nashville, TN, USA. March 2023.
5. *Mixture margin random-effects copula models for conserved microbial covariations.* JSM 2022. Washington, DC, USA. August 2022.
6. *Inference for temporally conserved microbial interactions using mixture margin copulas.* ENAR 2022. Houston, TX, USA. March 2022.
7. *Copula models for temporally conserved microbial interactions.* JSM 2021. Seattle, WA, USA (Virtual due to COVID-19). August 2021.
8. *Mixed margin copula models for microbial interactions.* ENAR 2021. Baltimore, MD, USA (Virtual due to COVID-19). March 2021.
9. *Zero-inflated topic models for the analysis of microbiome data.* ENAR 2020. Nashville, TN, USA. (Virtual due to COVID-19). March 2020.

Teaching

University of Pittsburgh

Fall 2026 PUBHLT 0411: Statistical Packages for Public Health Data Analysis (3 credits)
- 2026: 6 enrolled

Fall 2026
Fall 2025
Fall 2024 BIOST 2143: Longitudinal and Clustered Data Analysis (3 credits)
- 2026: 35 enrolled
- 2025: 32 enrolled
- 2024: 37 enrolled

Spring 2024
Fall 2023 BIOST 2025: Biostatistics Seminar (1 credit)
- 2024: 22 enrolled
- 2023: 17 enrolled

University of Pennsylvania

Teaching Assistant

Fall 2020 BSTA 630: Statistical Methods and Data Analysis I
Spring 2020 BIOM 611: Statistical Methods for the Design and Analysis of Experiments

Columbia University

Teaching Assistant

Fall 2017 P8130: Biostatistical Methods I

Other teaching

Spring 2018 P8180: Relational Databases & SQL Programming for Research & Data Science (Curriculum developer)

Advising

Doctoral dissertation advising

2025-Present 1. Tianshu Liu, PhD in Biostatistics, University of Pittsburgh

2024-2026 2. Crystal Zang, PhD in Biostatistics, University of Pittsburgh (Co-advised with Lu Tang)
- Miheala Serban Award for Best Poster Presentation - ASA Pittsburgh Chapter 2024 Banquet
- Best Poster Award, ASA Section on Text Analysis - JSM 2025
- Honorable Mention, ASA Mental Health Statistics Section - JSM 2026
- **First position:** Senior Statistician, Johnson & Johnson

Master's thesis advising

2026-2027 1. Hannah Schilpp, MS in Biostatistics, University of Pittsburgh
- 2026 Best MS Comprehensive Exam Award

2026 2. Ryan Karbon, MS in Biostatistics, University of Pittsburgh

2025-2026 3. Ben Witkowski, MS in Biostatistics, University of Pittsburgh
- 2026 Delta Omega Membership
- **First position:** Health Science Specialist, VA Pittsburgh Healthcare System

2024-2025 4. Nicholas Heaton, MS in Biostatistics, University of Pittsburgh
- 2025 Delta Omega Membership
- **First position:** Intermediate Statistician, University of Michigan

GSR advising

2026-Present 1. Xueqi (Ella) Shen (PhD GSR), University of Pittsburgh

2026-Present 2. Hannah Schilpp (MS hourly), University of Pittsburgh

2025-2026 3. Crystal Zang (PhD GSR), University of Pittsburgh

2024-Present 4. Tianshu Liu (PhD GSR), University of Pittsburgh

2024-2025 5. Kangyi Zhao (PhD GSR), University of Pittsburgh

2024-2025 6. Nicholas Heaton (MS hourly), University of Pittsburgh

Undergraduate advising

2026 1. Hagen Kalynko, BS in Public Health, University of Pittsburgh (PADOH field placement)

2025-Present 2. Patrick Gillespie, BS in Public Health, University of Pittsburgh

2025-2026 3. Monica Chelsea Seegars, BS in Statistics and Economics, University of Pittsburgh

2024-2025 4. Gala Krsmanovic, BS in Cyberpsychology, New Jersey Institute of Technology

High school advising

2024 1. Nina Tommarello

Other advising

2026-Present 1. Niloufar Dousti Mousavi, T32 Postdoctoral scholar, University of Pittsburgh (Scholarship Oversight Committee member)

2024-Present 2. Arsia Jamali, Gastroenterology fellow, University of Pennsylvania

Academic advisor

2026-Present 1. Xueqi (Ella) Shen, PhD in Biostatistics, University of Pittsburgh
2026-Present 2. Monica Chelsea Seegars, MS in Biostatistics, University of Pittsburgh
2025-2026 3. Ben Witkowski, MS in Biostatistics, University of Pittsburgh
2025-2026 4. Vani Patel, MS in Biostatistics, University of Pittsburgh
2025-2026 5. Crystal Zang, PhD in Biostatistics, University of Pittsburgh
2024-Present 6. Tianshu Liu, PhD in Biostatistics, University of Pittsburgh
2024-2025 7. Ishta Madan, MS in Biostatistics, University of Pittsburgh
2024-2025 8. Nicholas Heaton, MS in Biostatistics, University of Pittsburgh

Doctoral dissertation committee served

2026-Present 1. Alex Grant, PhD in Epidemiology, University of Pittsburgh
2026-Present 2. Mengyi Li, PhD in Epidemiology, University of Pittsburgh
2026-Present 3. Ying-Ju Lai, PhD in Biostatistics, University of Pittsburgh
2026-Present 4. Chiraag Gohel, PhD in Biostatistics, George Washington University
2026-Present 5. Sarah Wang, PhD in Biostatistics, University of Pittsburgh
2025-2026 6. Penghui Huang, PhD in Biostatistics, University of Pittsburgh
2025-Present 7. Xinliei Chen, PhD in Biostatistics, University of Pittsburgh
2025-2026 8. Rick Chang, PhD in Biostatistics, University of Pittsburgh
2024-Present 9. Emma Gay, PhD in Epidemiology, University of Pittsburgh
2024-2025 10. Jie Sun, PhD in Human Genetics, University of Pittsburgh
2024-2025 11. Wenjia Wang, PhD in Biostatistics, University of Pittsburgh
2023-Present 12. Preeti Bhanap, PhD in Computational Biology, University of Pennsylvania

Master's thesis committee served

2026-Present 1. Anastasia Pasynkov, MPH in Public Health Genetics, University of Pittsburgh
2026 2. Vani Patel, MS in Biostatistics, University of Pittsburgh
2025-2026 3. Xinrui Xie, MS in Biostatistics, University of Pittsburgh
2025-2026 4. Grace Luikart, MS in Biostatistics, University of Pittsburgh
2025-2026 5. Caroline Layding, MS in Biostatistics, University of Pittsburgh
2025-2026 6. Vivian Zhang, MS in Biostatistics, University of Pittsburgh
2024-2025 7. Haoling Wang, MS in Biostatistics, University of Pittsburgh
2024-2025 8. Aparna Nigam, MS in Biostatistics, University of Pittsburgh
2024-2025 9. Samantha Pokorny, MS/MPH in Human Genetics, University of Pittsburgh

Professional service

Editorial service

Paper referee: Annals of Applied Statistics, Annals of Epidemiology, Bioinformatics, Biometrics, Biostatistics, BMC Medical Research Methodology, Briefings in Bioinformatics, JAMA Oncology, Journal of the American Statistical Association (JASA), Methods in Ecology and Evolution, Scientific Reports, Statistica Sinica

Departmental, school, and university service

2026 Mentor, Public Health Science Academy
2026-Present Chair, Biostatistics PhD Applied Qualifying Exam Committee
2026-Present Member, Biostatistics MS Comprehensive Exam Committee
2026 Judge, Biostatistics Research Day
2025-2026 Member, Biostatistics Faculty Search Committee
2025 Judge, Biostatistics Research Day
2024-Present Member, Biostatistics MS Program Committee
2024-Present Member, Biostatistics PhD Qualifying Exam Committee
2024-Present Reviewer, Psychiatry Research Review Committee
2024 Judge, Biostatistics Research Day
2024 Organizer, Biostatistics Research Day
2023-Present Member, Outreach and Community Engagement Committee

Memberships

2026-Present International Statistical Institute
2021-Present Institute of Mathematical Statistics
2019-Present American Statistical Association
2019-Present Eastern North American Region International Biometric Society (ENAR IBS)

National and international service

2024-Present Secretary, ASA Section on Statistics in Genomics and Genetics
2023-2024 Local organizing committee, STATGEN 2024, ASA Section on Statistics in Genomics and Genetics

Conference service

2027 Invited session organizer, JSM 2027 conference
2027 Invited session chair, ENAR 2027
2026 Topic contributed session organizer and chair, JSM 2026
2026 Invited session organizer, ISI Regional Statistics Conference 2026
2025 Student paper competition reviewer, ENAR 2026
2025 Invited session organizer, ENAR 2025
2024 Student paper competition reviewer, ENAR 2025
2024 Session moderator, Rust Belt Microbiome 2024
2024 Invited session organizer and chair, STATGEN 2024 ASA SSGG

Other service

2024 Mentor, NJIT Women with STEAM Mentoring